

Nathan R. Gooneratne

360 Huntington Ave, Boston, MA 02115 | gooneratne.n@northeastern.edu | 610-348-4303 | skyloh.itch.io | skyloh.github.io | github.com/Skyloh

EDUCATION

Northeastern University, Boston MA

Aug. 2022 - Present

Khoury College of Computer Sciences

Candidate for Bachelor of Computer Science and Game Development

Expected Graduation: December 2025

GPA 3.95/4.00 - Honors Program

Dean's List: All Semesters

Relevant Coursework: Object-Oriented Design, Algorithms, Game Artificial Intelligence, Computer Graphics, Building Game Engines, Intro to C++, Linear Algebra, Game Concept Development, Level Design and Game Architecture, Fundamentals of Computer Networks, Foundations of Psychology, Fundamentals of Game Design

COMPUTER KNOWLEDGE

Languages - Proficient: **C#, Java, C++, Python**; Familiar: **C, OpenGL, Swift, HTML, CSS, JavaScript, HLSL**

Software - Proficient: **Unity, Git, Visual Studio, VSCode, IntelliJ, Blender**; Familiar: **Plastic SCM, Godot, Unreal**

Systems - Proficient: **Windows, MacOS**; Familiar: **Linux**

PERSONAL AND ACADEMIC PROJECTS

- | | | |
|-------------------------------|---|------------------------------|
| Oil City:
(Academic) | <ul style="list-style-type: none">Developed a Unity city-builder resource-management game with 9 other students in a mock independent studio environmentImplemented a tree-based node algorithm for a system of distributing game ticks to active objects used by other programmersCollaborated with industry mentors and QA to refactor the pipe-placing mechanic to suit user feedback and design details | July 2024 - Aug. 2024 |
| Project Dall-E:
(Personal) | <ul style="list-style-type: none">Created Dall-E-inspired first-person puzzle game with a custom asset importer and semantic-scoring algorithm to facilitate puzzle development processBuilt custom dialogue system with variable text speed and dialogue-world eventsProgrammed NPCs with Navmesh collision-avoidance roaming behaviors and custom animations with ShapeKeys and dynamic Rigging multi-aim constraints | Mar. 2023 - Apr. 2023 |

WORK EXPERIENCE

Northeastern University, Boston, MA

Sept. 2023 - Dec. 2023

Teaching Assistant for Object-Oriented Design

- Held office hours and lab 8 hours per week with other TAs to provide academic support for ~400 students regarding code debugging, feedback on student implementations, and explanations of design patterns and concepts
- Assisted in professorial tasks like exam proctoring, grading ~130 student assignments, and evaluating final projects
- Mentored students in proper OoD design patterns in the context of **Java** in **IntelliJ** with **Git** source control

Gamer Realities LLC

Sept. 2024 - Dec. 2024

Multiplayer VR Programmer

- Developed an online multiplayer physics-based chatroom with **Photon Fusion 2** as a full-time Unity programmer
- Utilized Unity editor backend functionalities to create developer tools to facilitate bug-fixing and implementation
- Implemented abstract and extendable systems for managing player status and ownership of network objects
- Collaborated with other developers through **Plastic SCM**, **HacknPlan**, and code-reviews for completed tasks

EXTRACURRICULAR ACTIVITIES

Game Studio Club @ Northeastern

Sept. 2022 - Present

- Produced 10 games as part of Game Jams or Long-Term Projects in teams with 3-5 other students
- Coordinated group tasks and made teaching materials/framework code documentation for Unity
- Elected Vice President in April 2023

INTERESTS

Drawing, D&D, Writing, Badminton, Animating, Swiss-German Language, Reading Science-Fiction/Fantasy, Baking